Feature-Based 2D Registration Evaluation

Accuracy and robustness summary from a configuration sweep on the synthetic demo pair.

Item	Value
Runs	27
Successful	24
Success rate	88.9%
Corner error (px) min / median / max	0.474 / 1.163 / 3.141
Mean inlier error (px) min / median / max	0.933 / 1.277 / 1.724

Experimental setup

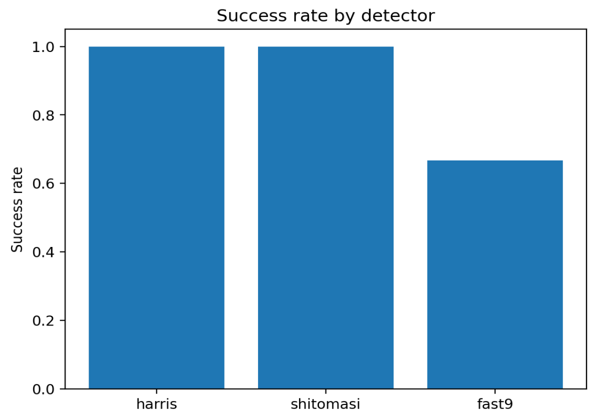
The evaluation uses the built-in synthetic image pair generated by the C++ demo. A full-factorial sweep was executed across 3 detectors, 3 descriptors, and 3 motion models (27 runs). All runs used the same matching and RANSAC settings: ratio=0.75, iters=700, mutual=1, symmetric=1, PROSAC=1, and RANSAC threshold=3.2 px.

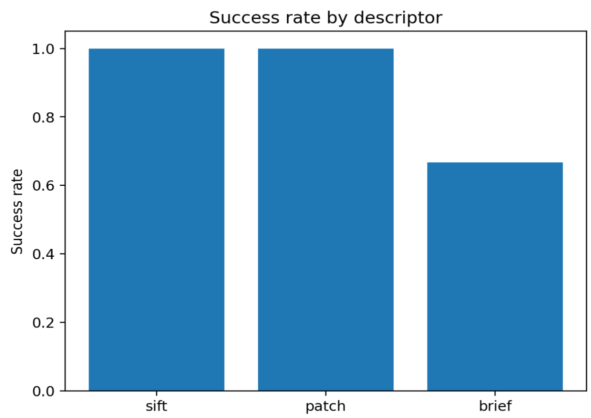
Metrics

Corner error (px) measures how well the estimated transform maps image corners compared to the known ground truth. **Mean inlier reprojection error (px)** is the average residual over inlier matches. **Success** indicates the solver produced a valid model (exit code 0).

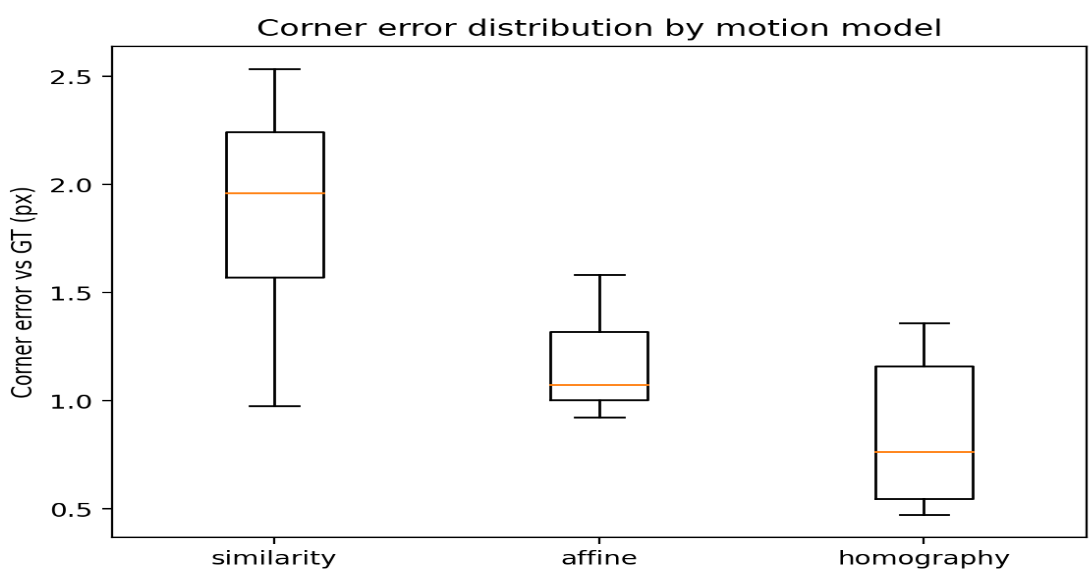
Results overview

Success rate





Corner error distribution by motion model



Aggregate results

Aggregate by model

Model	Runs	Success	Corner med	Corner range	Mean inlier err med
affine	9	88.9%	1.075	0.923–1.851	1.432
homography	9	88.9%	0.764	0.474–3.141	1.277
similarity	9	88.9%	1.962	0.977–2.535	0.969

Aggregate by detector and descriptor

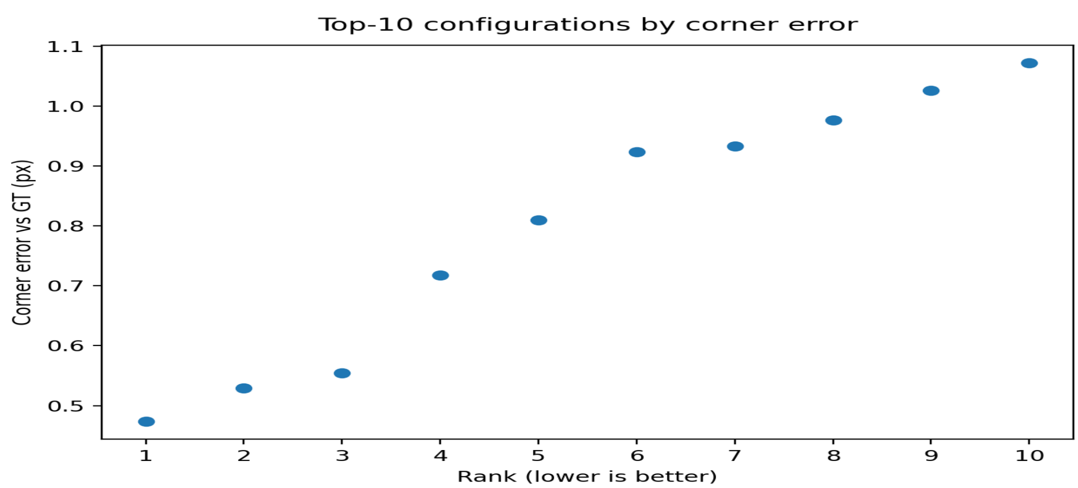
Detector	Runs	Success	Corner med
fast9	9	66.7%	2.193
harris	9	100.0%	0.933
shitomasi	9	100.0%	1.094

Descriptor	Runs	Success	Corner med
brief	9	66.7%	1.075
patch	9	100.0%	1.361
sift	9	100.0%	1.276

Top configurations

Top-10 ranked by corner error (lower is better).

Detector	Descriptor	Model	Matches	Inliers	Inlier ratio	Mean inlier err	Corner err
harris	brief	homography	28	28	100.0%	1.143	0.474
harris	patch	homography	20	19	95.0%	1.109	0.529
harris	sift	homography	28	25	89.3%	1.152	0.555
shitomasi	patch	homography	43	40	93.0%	1.276	0.718
shitomasi	sift	homography	48	46	95.8%	1.278	0.810
harris	sift	affine	28	24	85.7%	0.988	0.923
harris	patch	affine	20	20	100.0%	1.296	0.933
harris	brief	similarity	28	6	21.4%	1.064	0.977
shitomasi	patch	affine	43	41	95.3%	1.449	1.026
shitomasi	brief	affine	41	41	100.0%	1.541	1.072

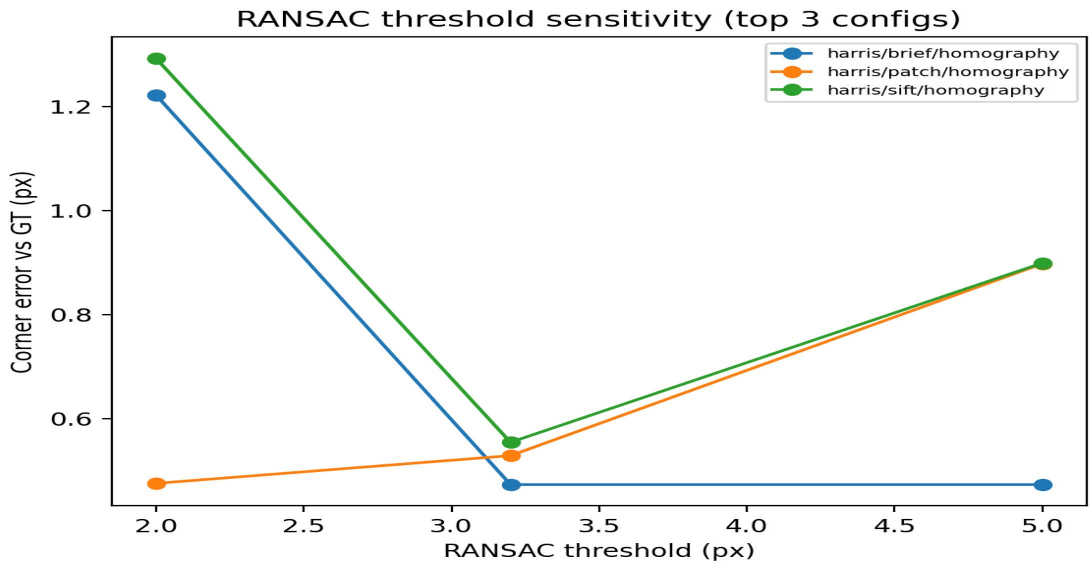


Observed failures

Detector	Descriptor	Model	Exit code
fast9	brief	similarity	2
fast9	brief	affine	2
fast9	brief	homography	2

RANSAC threshold sensitivity

Three best configurations were re-run at thresholds 2.0 / 3.2 / 5.0 px.



Detector	Descriptor	Model	Thresh (px)	Matches	Inliers	Inlier ratio	Corner err (px)
harris	brief	homography	2.0	28.0	24.0	85.7%	1.222
harris	brief	homography	3.2	28.0	28.0	100.0%	0.474
harris	brief	homography	5.0	28.0	28.0	100.0%	0.474
harris	patch	homography	2.0	20.0	16.0	80.0%	0.476
harris	patch	homography	3.2	20.0	19.0	95.0%	0.529
harris	patch	homography	5.0	20.0	20.0	100.0%	0.898
harris	sift	homography	2.0	28.0	21.0	75.0%	1.292
harris	sift	homography	3.2	28.0	25.0	89.3%	0.555
harris	sift	homography	5.0	28.0	26.0	92.9%	0.899

Recommendations from this run

Homography produced the lowest corner errors for the leading configurations, consistent with a synthetic transform that includes perspective components. Harris and Shi-Tomasi were stable across all descriptors. FAST9 paired with BRIEF was brittle in this sweep. RANSAC threshold selection had a measurable effect on corner error and depends on descriptor choice.